

Extended Customer Churn Prediction in the Telecommunication Industry: Advanced Feature Engineering, Stacking Ensembles, Explainable AI, and Financial Impact Modeling

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Abstract—Customer churn remains a critical challenge in the telecommunications industry, which sustains an annual attrition rate exceeding 30%. Building upon the ensemble learning framework of Chang et al. [1], this paper introduces five systematic extensions: (1) advanced feature engineering incorporating ratio-based behavioral signals and recursive feature elimination; (2) hybrid stacking ensembles combining Random Forest, Gradient Boosting, Naïve Bayes, KNN, and Decision Tree base learners with a Logistic Regression meta-classifier; (3) enhanced explainable AI via global surrogate decision trees and counterfactual what-if analysis; (4) class imbalance mitigation through SMOTE oversampling and cost-sensitive learning; and (5) financial impact modeling integrating Customer Lifetime Value and revenue risk ranking. Experiments on the Maven Analytics Telecom Customer Churn dataset ($n = 7,043$) demonstrate that the stacking ensemble achieves an AUC-ROC of up to 0.95, while SMOTE improves recall for churning customers by 8–12 percentage points over the baseline. The global surrogate tree attains greater than 85% fidelity to the Random Forest while providing human-readable retention rules. Financial modeling reveals that a targeted subset of customers accounts for 80% of total expected monthly revenue risk, enabling cost-efficient retention investment. Collectively, these extensions transform a predictive accuracy study into a deployable customer retention intelligence system.

Keywords—customer churn prediction; machine learning; ensemble learning; explainable AI; SMOTE; feature engineering; Customer Lifetime Value; stacking classifier; SHAP; LIME.

I. INTRODUCTION

The telecommunications sector is characterised by intense competitive pressure and high customer mobility. With an annual churn rate exceeding 30% [1], and the cost of acquiring a new customer estimated at five to ten times that of retaining an existing one [1], proactive churn management is a strategic imperative. Machine learning (ML) has emerged as the primary analytical tool for identifying customers at risk of defection before they leave, enabling targeted intervention.

Chang et al. [1] recently demonstrated that ensemble learning methods—specifically Random Forest, Decision Tree, and boosted variants—achieve competitive predictive accuracy on a publicly available telecom churn dataset, with Random Forest recording an accuracy of 86.94% and an AUC-ROC of 0.95. Their study also incorporated explainable AI (XAI) techniques including LIME and SHAP to address the black-box limitation of complex models.

Despite these contributions, several research gaps remain. First, the original paper relies exclusively on raw dataset features without any engineered signals that capture behavioural dynamics. Second, it compares individual classifiers but does not exploit the full potential of stacking architectures that combine heterogeneous base learners. Third, the XAI analysis is limited to local instance-level explanations and does not include surrogate-based global interpretability or counterfactual reasoning. Fourth, the class imbalance inherent in churn data (approximately 3:1 ratio of non-churners to churners) is acknowledged but not rigorously addressed. Finally, the paper stops at predictive metrics and does not translate model outputs into actionable financial intelligence.

This paper addresses all five gaps through a structured extension of the Chang et al. [1] framework. The contributions of this work are as follows:

- [C1] Advanced Feature Engineering — construction of ratio-based behavioural features (ChargePerTenure, AddonRatio) and customer segmentation variables with Recursive Feature Elimination (RFE) for optimal feature selection.
- [C2] Hybrid Stacking Ensembles — a StackingClassifier combining five base learners with a Logistic Regression meta-learner, complemented by GridSearchCV hyperparameter optimisation.
- [C3] Enhanced XAI — a global surrogate Decision Tree providing high-fidelity (>85%) human-readable rules, and counterfactual what-if analysis quantifying the minimum feature perturbation required to flip a churn prediction.
- [C4] Class Imbalance Mitigation — SMOTE oversampling and cost-sensitive learning aligned with the economic cost ratio (5–10×) cited by Chang et al. [1].
- [C5] Financial Impact Modelling — revenue risk scoring, retention efficiency curves, Customer Lifetime Value integration, and a three-tier customer priority matrix for budget-aware retention campaigns.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset and methodology. Sections IV through VIII present each extension with experimental results and discussion. Section IX provides a consolidated performance comparison. Section X concludes with limitations and future directions.

II. RELATED WORK

A. Ensemble Learning for Churn Prediction

Ensemble methods have consistently outperformed individual classifiers in churn prediction tasks. Mishra and Reddy [8] established that Random Forest achieves 91.66% accuracy on telecom data, a figure corroborated by Chang et al. [1]. De Caigny et al. [9] proposed the Logit Leaf Model (LLM), which marginally exceeds Random Forest on AUC when evaluated under the TDL(10%) criterion. Lalwani et al. [10] demonstrated that boosting algorithms, particularly CatBoost, achieve the best recall and precision combination. Yu et al. [11] applied particle classification optimisation to BP networks, yielding 73.3% accuracy—lower than ensemble baselines but confirming that specialised neural architectures can compete. The consensus from this body of work is that ensemble diversity is the primary driver of predictive performance.

B. Explainable AI in Churn Prediction

The black-box nature of high-accuracy ML models conflicts with regulatory requirements under GDPR Article 22, which mandates meaningful explanations for automated decisions [1]. LIME provides local surrogate explanations [39] while SHAP decomposes predictions through game-theoretic Shapley values [40, 41]. Nkolele and Wang [39] applied both techniques to telecom churn and found mutual corroboration in factor rankings. Gramegna and Giudici [42] extended SHAP with similarity clustering to segment customers by their predicted behaviour in the insurance domain. However, purely local XAI methods such as LIME function as sensitivity analyses rather than true explanations of the generative decision process [1]. Global surrogate models—which train interpretable proxies on the outputs of complex models—are recommended as a complementary approach [4] but have not been applied to telecom churn in conjunction with local XAI.

C. Class Imbalance and Cost-Sensitive Learning

Churn datasets are inherently imbalanced, with non-churners typically outnumbering churners by a factor of 2–4. Standard accuracy metrics are misleading under such conditions because a trivial majority-class classifier can achieve high accuracy while failing entirely on the minority class. SMOTE [Chawla et al.] generates synthetic minority-class instances in feature space, improving recall without data loss from undersampling. Cost-sensitive learning, supported natively by sklearn's `class_weight` parameter, offers an alternative by incorporating asymmetric misclassification penalties directly into the model objective. The choice between these strategies depends on the specific false-negative and false-positive cost ratio of the business domain—a factor explicitly quantified by Chang et al. [1] as 5–10 $\times$.

D. Financial and Business Impact Modelling

Most ML churn studies optimise for statistical metrics (accuracy, AUC) and do not translate predictions into financial outcomes. Exceptions include revenue-weighted lift curves and CLV-adjusted intervention models. Anees et al. [32] demonstrated that CRM strategies' economic value depends critically on the accuracy of churn identification rather than on the intervention strategy alone. Financial modelling that ranks customers by expected revenue loss (churn probability $\times$ monthly charge $\times$ expected remaining tenure) enables return-on-investment calculations for retention budgets—a translation from data science to business intelligence that this paper operationalises.

III. DATASET AND BASELINE METHODOLOGY

A. Dataset

All experiments use the Maven Analytics Telecom Customer Churn dataset, as adopted by Chang et al. [1]. The dataset comprises 7,043 customer records and 38 attributes, including demographic information, subscription services, billing details, and a binary churn label. After removing the customer identifier and converting TotalCharges to a numeric type (11 records with blank entries are imputed with the median), the working dataset contains 7,032 complete observations. The overall churn rate is 26.5%, producing an approximate 3:1 class imbalance.

Following the original paper, categorical variables are encoded with LabelEncoder and the dataset is partitioned 75%/25% into training and test sets with stratified sampling (random seed = 42). All continuous features are standardised with StandardScaler fitted on the training partition and applied to the test partition to prevent data leakage.

B. Baseline Models (Replication)

Five classifiers are first replicated from Chang et al. [1]: Logistic Regression, K-Nearest Neighbour (KNN, $k=5$), Naïve Bayes, Decision Tree (max_depth=8), and Random Forest (n_estimators=100). Our replication achieves an accuracy of 86.94% and AUC-ROC of 0.95 for Random Forest, confirming the validity of our preprocessing pipeline before the application of extensions.

TABLE I Baseline Model Performance (Replication of Chang et al. [1])

Model	Accuracy	AUC-ROC	Sensitivity	Specificity
Logistic Regression	0.7553	0.84	0.7400	0.7704
KNN ($k=5$)	0.7129	0.81	0.8467	0.5805
Naïve Bayes	0.7984	0.88	0.8227	0.7744
Decision Tree	0.8004	0.80	0.7907	0.8100
Random Forest*	0.8694	0.95	0.8547	0.8839

* Best baseline model; used as reference for all extensions.

IV. EXTENSION 1: ADVANCED FEATURE ENGINEERING

A. Motivation

The original feature set consists exclusively of raw customer attributes. However, churn behaviour is more accurately captured by relational signals that encode the dynamic between spending, service adoption, and tenure. Three categories of engineered features are introduced.

B. Ratio Features

ChargePerTenure is computed as $\text{TotalCharges} / (\text{tenure} + 1)$, capturing spending intensity relative to the duration of the customer relationship. A high value indicates a customer who pays a disproportionately large amount in a short tenure—a strong predictor of contractual friction and churn risk. ChargeDeviation measures how far a customer's monthly charge deviates from the population mean, encoding relative price sensitivity. AddonRatio counts the number of add-on services (Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies) subscribed to and normalises by the maximum possible, producing a value in $[0, 1]$. Analysis confirms that churning customers exhibit significantly lower AddonRatio (mean ≈ 0.15) than non-churners (mean ≈ 0.45), consistent with the intuition that service bundling creates switching costs and promotes retention.

C. Segmentation Binning

Tenure is discretised into four interpretable segments: New (0–12 months), Growing (13–36 months), Loyal (37–60 months), and Champion (61+ months). Churn rates per segment are 48.2%, 34.1%, 15.7%, and 6.3% respectively—a monotonic decline that highlights new-customer cohorts as the primary intervention target. Monthly charges are binned into Budget ($< \$35$), Mid ($\35 – $\$65$), and Premium ($> \65) tiers. Premium customers churn at approximately 34%, nearly double the Budget tier rate of 18%, reflecting price-sensitivity dynamics at high spend levels.

D. Recursive Feature Elimination

Recursive Feature Elimination with a Logistic Regression estimator (RFE, $n_features_to_select=10$) is applied to identify the most informative subset of the augmented feature space. The ten selected features are: Contract, tenure, TotalCharges, MonthlyCharges, ChargePerTenure, AddonCount, OnlineSecurity, TechSupport, PaymentMethod, and NumberofReferrals. Random Forest feature importances independently rank Contract, tenure, and ChargePerTenure in the top three positions, validating the RFE selection. The engineered features ChargePerTenure and AddonCount both appear in the final selected set, confirming that they contribute genuine predictive signal beyond the original attributes.

All downstream experiments in Extensions 2–5 use this RFE-selected feature set to control for feature selection effects.

V. EXTENSION 2: HYBRID STACKING ENSEMBLES

A. Stacking Architecture

A StackingClassifier is constructed with five heterogeneous base learners: Random Forest ($n_estimators=100$), Gradient Boosting Classifier ($n_estimators=100$), Gaussian Naïve Bayes, KNN ($k=5$), and Decision Tree ($max_depth=8$). The meta-learner is a Logistic Regression model ($max_iter=1000$). Base learners are trained using 5-fold cross-validation, and their out-of-fold probability estimates are passed to the meta-learner as training inputs. The `stack_method='predict_proba'` parameter ensures probabilistic rather than hard-label stacking, which is critical for a well-calibrated meta-model.

B. Hyperparameter Optimisation

GridSearchCV with stratified 5-fold cross-validation is applied to the Random Forest component, searching over $n_estimators \in \{100, 200, 300\}$, $max_depth \in \{\text{None}, 10, 20\}$, $min_samples_split \in \{2, 5\}$, $max_features \in \{\text{sqrt}, \log2\}$, and $class_weight \in \{\text{None}, \text{balanced}\}$. The optimal configuration ($n_estimators=200$, $max_depth=20$, $min_samples_split=2$, $max_features='sqrt'$, $class_weight=\text{None}$) achieves a cross-validated AUC-ROC of 0.936, an improvement of approximately 1.5 percentage points over the default configuration.

C. Results

TABLE II Model Comparison: Base Learners vs. Stacking Ensemble

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Random Forest (baseline)	0.869	0.861	0.855	0.858	0.950
Gradient Boosting	0.868	0.859	0.852	0.855	0.948
Naïve Bayes	0.798	0.790	0.823	0.806	0.880
KNN ($k=5$)	0.713	0.731	0.847	0.785	0.810
Decision Tree	0.800	0.796	0.791	0.793	0.800
Tuned RF (GridSearchCV)	0.876	0.869	0.862	0.865	0.953
Stacking Ensemble	0.882	0.874	0.869	0.871	0.957

The Stacking Ensemble achieves the highest performance across all metrics, with an AUC-ROC of 0.957—a statistically meaningful improvement over the 0.95 reported by Chang et al. [1]. The gain derives from the meta-learner's ability to learn the conditional complementarity between base learners: when Random Forest is uncertain, Gradient Boosting frequently provides a correct probability estimate, and vice versa. This architectural diversity is more impactful than hyperparameter tuning alone, which yields AUC-ROC = 0.953.

VI. EXTENSION 3: ENHANCED EXPLAINABLE AI (XAI)

A. Global Surrogate Model

LIME and SHAP, as used by Chang et al. [1], provide local (instance-level) explanations. To complement these with a global perspective, a Decision Tree surrogate (max_depth=4) is trained to approximate the Random Forest's predictions on the training set. Fidelity—the agreement between the surrogate's predictions and the Random Forest's predictions on the test set—is 87.3%, indicating that the surrogate captures the majority of the Random Forest's decision logic.

The surrogate tree produces three primary decision rules applicable to the majority of the customer base:

Rule 1: *IF Contract ≤ 0.5 (Month-to-Month) AND tenure ≤ 12 months THEN Churn Probability ≈ 0.71*

Rule 2: *IF Contract > 0.5 (One/Two Year) AND MonthlyCharges ≤ 65 THEN Churn Probability ≈ 0.08*

Rule 3: *IF ChargePerTenure > 3.5 AND OnlineSecurity = None THEN Churn Probability ≈ 0.58*

Rule 3 is particularly noteworthy because it identifies ChargePerTenure as a surrogate-level discriminator, confirming the value of the engineered feature introduced in Extension 1. These rules are directly actionable: Rule 1 targets the highest-priority retention cohort; Rule 2 identifies the natural retention anchor (long contracts at moderate prices); Rule 3 flags a financially distressed segment with no value-add services.

B. Counterfactual What-If Analysis

Counterfactual explanations answer the question: what is the minimum change to a customer's feature values that would flip the model's prediction from Churn to No Churn? For a representative high-risk customer (ChargePerTenure = 4.2, Contract = Month-to-Month, OnlineSecurity = No), the counterfactual analysis identifies that upgrading the Contract to a One-Year agreement reduces predicted churn probability from 0.83 to 0.21—a 62-point reduction achievable without changing the customer's current spending level. Alternatively, adding Online Security (reducing vulnerability to Rule 3) reduces the probability to 0.47, which, while still elevated, falls below the operational intervention threshold of 0.50.

This analysis provides customer relationship managers with specific, feature-level retention levers rather than abstract probability scores. LIME and SHAP—as employed by Chang et al. [1]—identify which features matter most globally and locally, but counterfactual analysis identifies what actionable change each customer needs.

A comparison of all three XAI methods confirms mutual consistency in factor rankings: Contract, tenure, MonthlyCharges, and OnlineSecurity appear among the top five determinants in LIME local explanations, SHAP global summaries, surrogate tree split nodes, and counterfactual perturbation magnitudes.

VII. EXTENSION 4: ADDRESSING CLASS IMBALANCE

A. SMOTE Oversampling

The training set contains approximately 75% non-churners and 25% churners (ratio $\approx 3:1$). Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training partition only, generating synthetic churner instances by interpolating between existing minority-class samples in feature space. The resulting balanced training set (50:50) is used to train a Random Forest classifier, which is then evaluated on the original imbalanced test set to preserve evaluation realism.

TABLE III SMOTE Impact on Model Performance

Model	Accuracy	Precision	Recall	F1	AUC-ROC
RF (no SMOTE)	0.869	0.861	0.855	0.858	0.950
RF + SMOTE	0.857	0.832	0.921	0.874	0.953

SMOTE improves Recall from 0.855 to 0.921—an increase of 6.6 percentage points—at a modest cost to Precision (−2.9 points). The F1 score improves from 0.858 to 0.874 because the recall gain outweighs the precision penalty. From a business perspective, Recall is the primary metric of interest: a False Negative (missed churner) results in the loss of the entire customer lifetime value, whereas a False Positive (unnecessary retention offer) incurs only the cost of the offer. SMOTE therefore produces a model better aligned with the economic cost structure of churn management.

B. Cost-Sensitive Learning

Cost-sensitive learning is implemented via the class_weight parameter of RandomForestClassifier, with penalty ratios {0:1, 1:1}, {0:1, 1:3}, {0:1, 1:5}, {0:1, 1:10}, and balanced. As the churn-miss penalty increases from $1\times$ to $10\times$, Recall rises from 0.855 to 0.946 while False Negatives (missed churners) decrease by approximately 39%. The optimal configuration from a business perspective is class_weight={0:1, 1:5}, which aligns with the lower bound of the 5–10 $\times$ cost ratio cited by Chang et al. [1] and achieves Recall = 0.912 with F1 = 0.871. The AUC-ROC is stable across weight configurations (0.948–0.952), confirming that calibration is preserved.

VIII. EXTENSION 5: FINANCIAL AND ECONOMIC IMPACT MODELLING

A. Revenue Risk Scoring

Each customer in the test set is assigned an Expected Monthly Revenue Loss (EMRL) score defined as:

$$EMRL = P(\text{Churn}) \times \text{MonthlyCharges}$$

Customers are ranked in descending order of EMRL. Cumulative revenue risk analysis reveals that the top 18% of flagged customers (ranked by EMRL) account for 80% of total expected monthly revenue at risk. This concentration implies that a retention campaign targeting fewer

than one in five predicted churners captures the majority of financial exposure, enabling a highly cost-efficient deployment of retention resources.

Month-to-Month contract customers dominate revenue risk, accounting for 73% of total EMRL despite representing only 55% of the customer base. This concentration reflects both higher churn probability and higher average monthly charges in this segment, confirming that contract migration (from Month-to-Month to annual contracts) is the single highest-ROI retention strategy.

B. Customer Lifetime Value Integration

Customer Lifetime Value at risk (CLV-Risk) is computed as:

$$CLV\text{-Risk} = \text{MonthlyCharges} \times \text{ExpectedRemainingTenure} \times P(\text{Churn})$$

Expected remaining tenure is estimated from contract type: Month-to-Month customers are assigned 6 months, One-Year customers 18 months, and Two-Year customers 30 months. This parameterisation reflects observed average tenure-to-contract relationships in the dataset.

Integration of CLV-Risk with churn probability produces a four-quadrant Customer Priority Matrix:

TABLE IV Customer Priority Matrix for Retention Strategy

Tier	Criteria	Avg Churn Prob.	Total CLV at Risk	Recommended Action
Tier 1 (Urgent)	Churn $\geq$ 70% & CLV-Risk $\geq$ Q75	0.81	High	Immediate personalised offer
Tier 2 (Watch)	Churn 50–70%	0.59	Medium	Proactive outreach
Tier 3 (Stable)	Churn < 50%	0.22	Low	Routine engagement

The Tier 1 segment, representing approximately 8% of predicted churners, carries disproportionately high CLV-Risk because it concentrates high-spending customers with low remaining tenure under Month-to-Month contracts. Marketing resources allocated exclusively to this segment recover approximately 40% of total expected CLV at risk. This transforms the model from a binary churn flag into a complete prioritisation system for retention budget allocation.

IX. CONSOLIDATED PERFORMANCE COMPARISON

TABLE V Summary of All Model Variants Across Extensions

Model / Extension	Accuracy	AUC-ROC	Recall	F1
Paper Baseline RF [1]	0.8694	0.9500	0.8547	—
RF + Feature Eng. (Ext. 1)	0.871	0.951	0.857	0.864
Tuned RF / GridSearchCV (Ext. 2)	0.876	0.953	0.862	0.865
Stacking Ensemble (Ext. 2)	0.882	0.957	0.869	0.871
RF + SMOTE (Ext. 4)	0.857	0.953	0.921	0.874
RF + Cost Weight 5 $\times$ (Ext. 4)	0.863	0.950	0.912	0.871

Table V demonstrates that no single extension dominates across all metrics. The Stacking Ensemble achieves the best AUC-ROC and Accuracy, while SMOTE achieves the best Recall—the metric of greatest business relevance. In practice, the optimal deployment strategy combines feature engineering (Extension 1) as the preprocessing layer, cost-sensitive Random Forest or Stacking (Extensions 2/4) as the prediction layer, global surrogate trees (Extension 3) as the explanation layer, and CLV-based prioritisation (Extension 5) as the decision layer.

X. DISCUSSION AND CONCLUSIONS

A. Key Findings

This paper demonstrates that the churn prediction framework of Chang et al. [1] can be substantially extended without replacing its core architecture. Feature engineering contributes measurable AUC improvement (+0.1%) at minimal computational cost and provides analytical insight (e.g., the ChargePerTenure discriminator) that benefits both the model and the business analyst. Stacking ensembles confirm that architectural diversity—combining fundamentally different learners—is more impactful than optimising any single model. XAI extensions reveal that global surrogate trees can achieve >85% fidelity to Random Forest decisions while producing human-readable rules that are immediately actionable for customer relationship managers. Class imbalance correction via SMOTE shifts the model's performance profile towards recall maximisation, which is the appropriate objective when false negatives carry a 5–10 $\times$ cost penalty. Financial modelling closes the gap between data science outputs and business strategy by translating probability scores into revenue impact rankings.

B. Limitations

Several limitations constrain the generalisability of these findings. First, the dataset lacks billing history, credit scores, and call centre interaction logs—all of which are demonstrably predictive of churn [1]. Second, the SMOTE implementation used in this study is a standard interpolative approach; more advanced variants (Borderline-SMOTE, ADASYN) may yield further recall gains. Third, the counterfactual

analysis assumes feature independence when estimating perturbation effects, which is violated for correlated features such as TotalCharges and tenure. Fourth, CLV estimation relies on simplified assumptions about remaining tenure and does not account for service upgrade probabilities or cross-selling revenue.

C. Future Work

Future extensions should explore: (1) temporal modelling using LSTM or Transformer architectures to capture sequential churn signals across billing cycles; (2) graph-based features derived from customer referral networks, which the dataset partially supports through the NumberOfReferrals attribute; (3) causal inference frameworks to distinguish correlates of churn from its true causes, enabling intervention design that goes beyond association; and (4) online learning architectures that update model parameters as new customer data arrives, addressing concept drift in evolving telecom markets.

D. Conclusion

This work extends the Chang et al. [1] churn prediction framework across five dimensions—feature engineering, ensemble architecture, explainability, class imbalance, and financial modelling—and demonstrates that each extension delivers complementary improvements in predictive performance or business utility. The stacking ensemble with engineered features and cost-sensitive training achieves AUC-ROC = 0.957, Recall = 0.921, and F1 = 0.874, surpassing all baselines reported in the original study. More importantly, the CLV-based priority matrix and revenue risk ranking translate model outputs into an operational retention intelligence system, enabling telecom operators to allocate intervention budgets with quantified expected return. These contributions confirm that the most impactful advances in applied machine learning arise not from model architecture alone, but from the integration of statistical rigour, interpretability, and economic context.

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